

Project Title: Predicting Drivers of Electric Vehicle Adoption Across U.S. States Using Socioeconomic Data

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1. ABSTRACT

Although electric vehicle (EV) adoption is expanding across the United States, rates vary widely among states due to underlying differences in socioeconomic conditions. This project examines whether three socioeconomic factors, bachelor attainment rate, labor force participation rate, and per capita income, can explain state-level variation in EV adoption and predict EV share (%) between 2016 and 2023. We modeled EV adoption rates using Multiple Linear Regression (MLR) and Random Forest regression using a cleaned dataset containing only essential socioeconomic variables and states. We hypothesized that states with higher educational attainment, stronger labor force participation levels, and higher income levels would lead to a higher rate of EV adoption. After applying a logarithmic transformation to EV share, the Random Forest model achieved an R^2 value of 0.6197 with per capita income as the strongest predictor. These results suggest that socioeconomic conditions play an important role in the adoption of EVs across states.

2. INTRODUCTION

2.1 Project Idea

EV adoption provides clear benefits for both climate change mitigation [1] and public health [2], yet these rates vary due to differences in socioeconomic conditions, infrastructure, and policy context. In this project, we aim to identify the socioeconomic factors that best explain EV adoption across states and forecast adoption rates for the next seven years. Our research question is: To what extent do socioeconomic indicators (bachelor attainment rate, labor force participation rate, and per capita income) predict EV share (%) across U.S. states between 2016 and 2023, and can these factors be used to project EV adoption?

We initially compared infrastructure and demographic data, but found that states with the highest numbers of charging stations were often the same states with the highest EV adoption rates. This

suggests that infrastructure tends to follow adoption, as the implementation of charging stations depends on the growth of EV demand. Infrastructure data correlates with population size, making it difficult to model fairly. Therefore, we pivoted to focus only on socioeconomic variables (education, income, and labor participation) as more consistent predictors of adoption.

We hypothesize that states are more likely to have high EV adoption rates (greater than 1 percent of total vehicles) if their working populations are highly college-educated and their per capita income exceeds the U.S. median (\$45,000) [3]. We use three thresholds to test this: 35 percent bachelor attainment rate, 75 percent labor force participation rate, and \$45,000 per capita income (Figure 1). These thresholds represent conditions associated with financial stability, employment participation, and the capacity to afford new technologies such as electric vehicles.

2.2 Dataset

File: EV_Data.csv (82kB), <https://www.kaggle.com/datasets/surajshivakumar/ev-adoption-usa>

The original dataset compiles metrics from multiple sources, including the National Renewable Energy Laboratory, U.S. Census Bureau, Energy Information Administration, and Bureau of Transportation. It contains over 500 records and more than 30 variables, including total EV registrations, EV share, education levels, income, population, and infrastructure indicators.

3. METHODOLOGY

3.1 Preprocessing

For this study, we began by reducing the original dataset from 36 variables to six columns: state, year, EV share (%), labor force participation rate (%), bachelor attainment rate (%), and per capita income (\$). We confirmed that the dataset contained no missing or duplicate values. In our initial assessment of outliers, we examined the distribution of all numeric variables using IQR-based diagnostics and visualization tools. The District of Columbia (D.C.) was a significant outlier, particularly in bachelor attainment rate and per capita income. D.C.'s attainment rate of 65.9 percent is substantially higher than the national average of 36.2 percent [4], reflecting its concentration of federal employment and atypical demographic composition. As a federal district rather than a state [5], D.C. is also uniquely urbanized and structurally different from the rest of

the dataset. We decided to remove D.C. from the modeling stage to maintain a more representative state-level sample.

3.2 Descriptive analysis

Our descriptive analysis examined how EV adoption and socioeconomic characteristics varied across U.S. states from 2016 to 2023. We began by summarizing each state's average bachelor attainment rate, labor force participation rate, and per capita income, and compared these values against our hypothesis thresholds. Our histogram visualizations showed wide variation in socioeconomic conditions across states, and only a subset of states met all three thresholds at the same time (Figure 1).

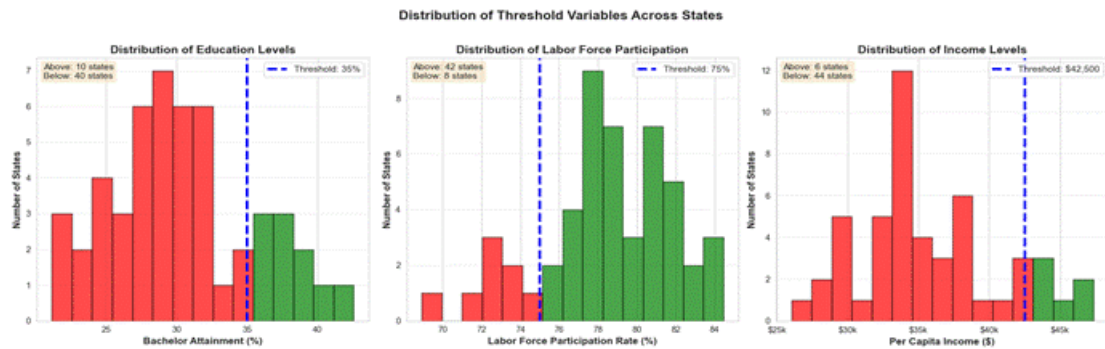


Figure 1. Thresholds used for testing the hypothesis

To explore temporal patterns between states, we plotted each variable across years and across states. EV share remained relatively low for most states, often below 1 percent, but increased steadily over the eight years. States such as California, Washington, and Oregon consistently appeared at the upper end of the EV distribution. The socioeconomic variables showed a much broader spread. Bachelor attainment and income levels varied strongly across states, while labor force participation showed clear regional patterns, with states such as West Virginia remaining consistently lower than the national average.

We also compared each socioeconomic variable directly to EV share using detailed scatterplots (Figure 2). The plots showed positive relationships between EV share and both bachelor's attainment and income. Income showed the strongest overall correlation with EV share at 0.669, while bachelor attainment produced a moderate positive correlation of 0.478.

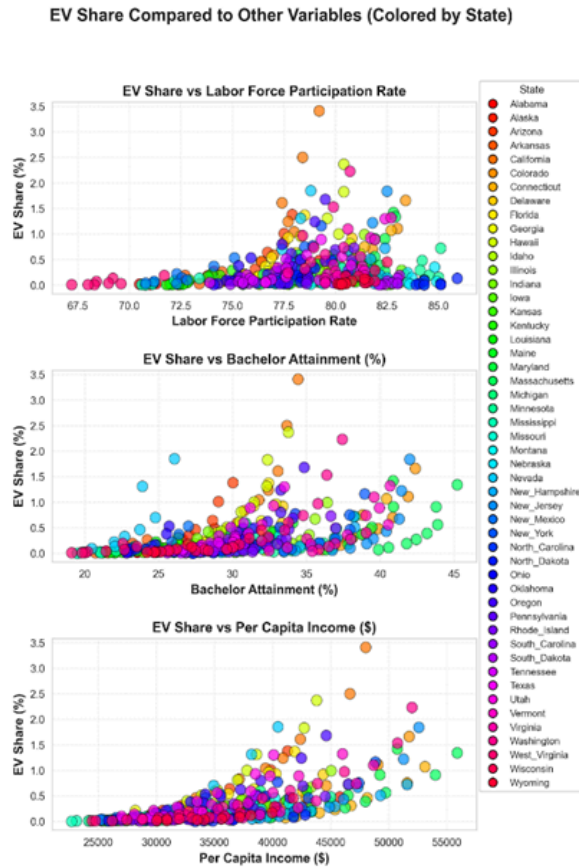


Figure 2. Scatterplots comparing EV share to labor force participation rate (top), bachelor attainment (middle), and per capita income (bottom).

3.3 Exploratory Data Analysis

For our exploratory data analysis, we began with univariate analysis. This included summary statistics for all numeric variables and producing histograms and boxplots to examine their distributions. This step allowed us to observe overall ranges, skewness, and variation in EV share, income, bachelor attainment, and labor force participation across states and years.

Next, we conducted bivariate analysis to assess how each socioeconomic factor related to EV share. We computed pairwise correlations for all numeric variables and visualized these using correlation heatmaps. Scatterplots were then created for EV share versus each numeric predictor.

However, several states deviated from these trends, such as Nevada, which combined moderate bachelor attainment with relatively high EV adoption. These exceptions suggested that socioeconomic indicators alone are not sufficient to fully explain adoption behavior. The curvature and spread observed in these scatterplots also indicated that a linear model would likely underfit these relationships, which motivated our transition to a Random Forest model that is better suited to capturing nonlinear trends. Finally, we calculated correlations for each socioeconomic variable against EV share, both overall and at the state level. These metrics, along with the scatterplot visualizations generated through the plotting code, helped confirm the direction and strength of each variable's association with EV adoption.

These visualizations helped identify the strength and direction of linear relationships and highlighted states that deviated from expected trends.

Additionally, we used the IQR method to confirm that states with extreme values, such as high income or unusually low labor participation, reflected real demographic conditions rather than data errors. Multivariate analysis was then conducted through an expanded correlation matrix, pair plots of key variables, and state-level comparisons. Pairwise visualizations with state-level hue highlighted how socioeconomic characteristics cluster geographically. We also examined EV adoption trajectories by plotting each state's EV share over time.

Finally, we segmented states into groups based on income and education terciles to compare mean EV adoption across socioeconomic tiers. We also created a normalized comparison of all key variables using z-scores to evaluate how each state's socioeconomic profile aligned with its average EV share. A heatmap of EV share by state and year provided an additional overview of temporal patterns.

3.4 Predictive Model Design

We initially attempted to model EV adoption using multiple linear regression, with EV share (%) as the dependent variable and bachelor's attainment, labor force participation, and per capita income as predictors. However, the linear regression model struggled to account for differences between low- and high-adoption states, producing lower R^2 values and higher prediction errors due to the nonlinear nature of the underlying relationships.

Instead, we used a Random Forest Regression model, which better models nonlinear patterns and interactions [6]. Random Forests were selected because they inherently capture nonlinear relationships more effectively than linear models. The dataset was split into an 80% training set and a 20% testing set, and because EV share is heavily right-skewed, we applied a log transformation to the target variable to stabilize variance and reduce the influence of high-adoption states. The Random Forest model was trained using the same three socioeconomic predictors.

3.5 Work Division

Our work division continues to be the same, as we all contributed to this project. Giselle worked on data preprocessing, regression modeling, and visualization. Daphne worked on data cleaning, exploratory analysis, and results visualization. Jose worked on feature selection, domain research, and predictive modeling.

4. MODEL RESULTS

4.1 Model Performance

We evaluated two predictive models, a linear regression model with a log-transformed target and a Random Forest model with tuned hyperparameters. The linear regression served as a baseline for comparison, while the Random Forest captured nonlinear relationships expected in EV adoption patterns.

	Linear Regression	Random Forest
MAE	0.1762	0.1632
RMSE	0.0754	0.0719
R ²	0.6012	0.6197

Table 1. Comparison of test metrics for linear regression and random forest models

Overall, the Random Forest improved on the baseline model by reducing prediction error and increasing explained variance (Table 1). The difference points to the fact that EV adoption patterns are not well approximated by a linear relationship and benefit from a model capable of learning nonlinear effects and interactions.

4.2 Model Results

The visual evaluation of the Random Forest model shows that the predictions align well with actual EV share values (Figure 3). In the actual vs predicted plot, most points fall close to the diagonal reference line, although the model tends to underpredict the highest-adoption states. The residuals versus predicted values reveal a mostly random distribution around zero, which indicates that the model does not exhibit strong bias across the prediction range. The residual histogram also shows a distribution centered near zero, with most errors falling within a narrow band. This supports the conclusion that the model is stable and performs consistently across states.

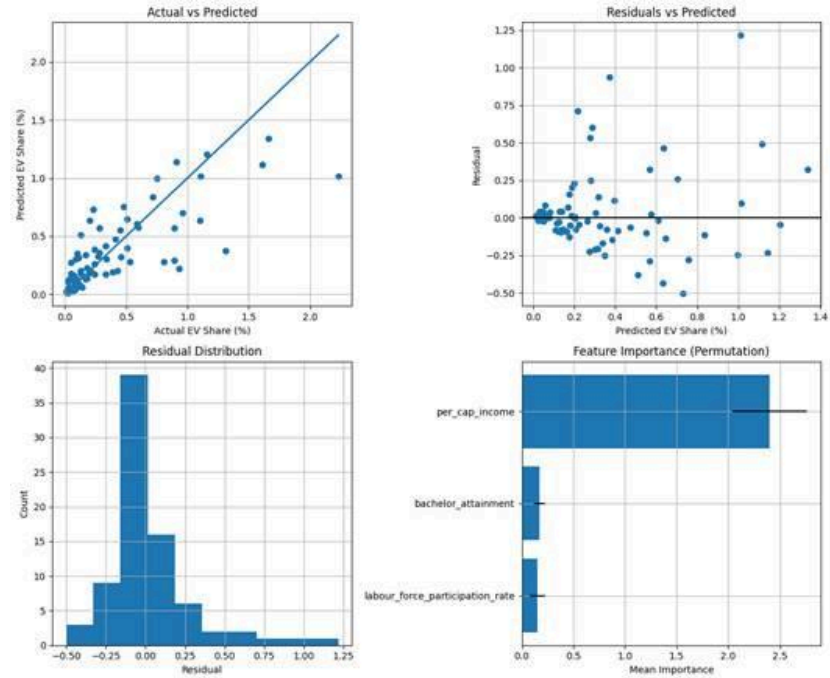


Figure 3 Random Forest model evaluation results.

Permutation feature importance shows a clear ranking of predictor influence.

- Per capita income: 2.40075 (± 0.36173)
- Bachelor attainment: 0.17043 (± 0.05188)
- Labor force participation: 0.14822 (± 0.07434)

Income stands out as the dominant predictor by a wide margin, indicating that economic capacity strongly influences EV adoption. Education and labor force participation contribute far less, though their nonzero scores still support the idea that more educated and economically engaged populations are more likely to adopt new technologies.

5. DISCUSSION AND CONCLUSIONS

The results show that the Random Forest provides a reasonable approximation of EV adoption across states. The model identifies per capita income as the strongest driver of EV adoption, highlighting the important role that economic capacity plays in this transition. States that have a higher income provide residents with greater financial means to purchase electric vehicles, which have a higher upfront cost than internal combustion engine vehicles [7]. The limited ability to afford these costs inhibits residents in lower-income states from adopting EVs at the same rate,

which aligns with the model's results. It is important to mention that although education and labor force participation played an important role in EV adoption, their roles were less influential relative to income.

Although the model performed well, socioeconomic factors don't fully explain differences in EV adoption. Accessibility to charging stations and fuel prices play important roles that were not captured by the model. Nevada is an important example, having high EV adoption rates despite having moderate socioeconomic indicators. This suggests that policy and infrastructure factors can influence adoption beyond education, income, or labor participation. Understanding these relationships is important for developing effective EV adoption strategies.

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